KERALA POLYTECHNIC · REVISION 2026

Electric Vehicle Technology & Design Considerations

Complete student lesson for course code **5411**.

Revision 2026 Semester 5 Program Core 4 modules

Course snapshot

Scheme: 4 (L:3 T:1 P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 60

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

5411

Semester

5

Credits

4

Teaching scheme

4 (L:3 T:1 P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Electric and Hybrid Vehicle Fundamentals	15	CO1
2	EV Control Systems, Drive Trains and Braking	15	CO2
3	EV Design Requirements, Dynamics and Motors	15	CO3
4	EV Braking, Sensors, Cooling, Service and Safety	15	CO4

Prerequisites

- 2411 — Introduction to Electric Vehicles & Elementary Theory, semester 2; prerequisite topic: basic knowledge about storage batteries.
- 3411 — Electric Vehicle Batteries & Charging Systems, semester 3; prerequisite topic: basic knowledge of batteries and charging systems.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To understand general aspects of Electric and Hybrid Vehicles.
2. To understand the control system basics of EVs.
3. To understand basic design considerations of EVs.
4. To understand the basic concepts of braking systems and sensors.

Course Outcomes

1. CO1: Outline the Electric Vehicle and Hybrid Vehicle — 15 hours — Understanding.
2. CO2: Identify the various drive trains in EV & HEV — 15 hours — Understanding.
3. CO3: Outline the basic design considerations of EV — 15 hours — Remembering.
4. CO4: Outline the braking system for EVs — 13 hours — Understanding.

CO-PO mapping as supplied

CO1: PO1 = 2 (Moderately mapped).

CO2: PO1 = 2 (Moderately mapped).
CO3: PO1 = 2 (Moderately mapped).
CO4: PO1 = 2 (Moderately mapped).
Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Need for hybrid and electric vehicles	4
module-1	M1.02	Components and working principles of electric and hybrid vehicles	4
module-1	M1.03	Configurations of Electric Vehicles	3
module-1	M1.04	Types of hybrid electric vehicle	4
module-2	M2.01	Control systems for EVs	4
module-2	M2.02	Split power devices for HEV	3
module-2	M2.03	Drive trains in EV and HEV	4
module-2	M2.04	Braking systems in EV	4
module-3	M3.01	Design requirements for electric vehicles	4
module-3	M3.02	Vehicle dynamics and performance	4
module-3	M3.03	Classification and requirements of EV motors	4
module-3	M3.04	Comparison of motors used in EVs	3
module-4	M4.01	Braking systems of EVs	3
module-4	M4.02	Regenerative braking and hybrid braking	4
module-4	M4.03	Sensors used in EVs	4
module-4	M4.04	Service and maintenance of EV components	4

Learning Roadmap

1. Electric and Hybrid Vehicle Fundamentals
CO1 · 15 hours

2. EV Control Systems, Drive Trains and Braking
CO2 · 15 hours

3. EV Design Requirements, Dynamics and Motors
CO3 · 15 hours

4. EV Braking, Sensors, Cooling, Service and Safety
CO4 · 15 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Electric and Hybrid Vehicle Fundamentals

This module establishes why electric and hybrid propulsion systems are used, how their energy and power paths are arranged, and how the principal vehicle configurations differ.

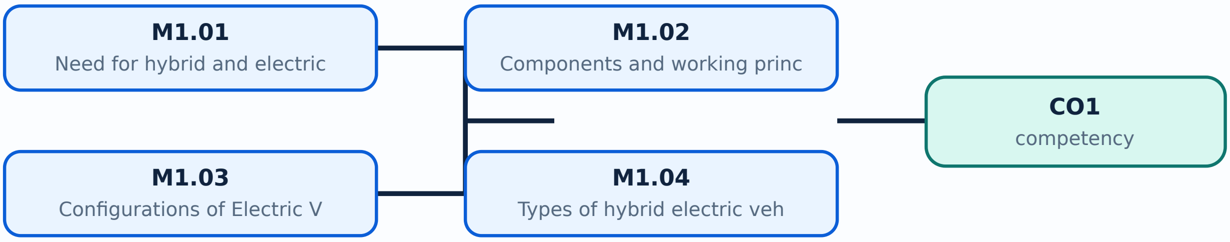
Hours: 15
CO1 · Bloom level: Understanding

Learning goals

- Relate the need for EVs and HEVs to energy, environmental, and transport requirements.
- Identify the main components and trace the working principle of electric and hybrid vehicles.
- Differentiate BEV, FCEV, HEV, PHEV, EREV, and common HEV configurations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Need for hybrid and electric vehicles (4 hours)

Concept and explanation

Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and the possibility of using electricity from diverse sources. Historical development moved from early battery vehicles through modern power electronics, lithium-based batteries, charging infrastructure, and electronically controlled propulsion. Social importance includes quieter transport and reduced local pollution; environmental benefit depends on the electricity-generation mix, battery manufacture, and responsible end-of-life management.

Malayalam support: EV, HEV □□□□□□□□□□ □□□□□□□ energy efficiency, emission reduction, quieter operation, regenerative braking □□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□. Environmental benefit □□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□□□; electricity source □□ battery life-cycle □□ □□□□□□□□□□□.

Study points

- State at least three technical or environmental reasons for adopting EVs or HEVs.
- Distinguish local tail-pipe emissions from total life-cycle environmental impact.
- Mention the role of batteries, motors, power electronics, and charging infrastructure.

Engineering / workplace application: Electric mobility is used in passenger cars, buses, delivery vehicles, forklifts, and two-wheelers where efficient stop-start operation is valuable.

Illustrative example: If a vehicle recovers 20 kJ during braking instead of dissipating it as heat, that recovered energy can be returned to the battery through the power electronics and motor-generator path.

Common mistake: Claiming that every EV has zero total environmental impact ignores electricity generation and battery manufacturing.

M1.02 — Components and working principles of electric and hybrid vehicles (4 hours)

Concept and explanation

A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loads, and vehicle control unit. An HEV adds an engine, fuel system, generator or motor-generator, and a supervisory energy-management controller. During propulsion, stored electrical energy passes through the inverter to the motor; during regenerative deceleration, the power flow reverses and the motor operates as a generator.

Malayalam support: Battery, inverter, traction motor, controller, DC-DC converter, charger, differential □□□□□□□□□□ EV-□□□□ □□□□□□ parts. HEV-□ engine □□□□ □□□□□□□□□. Propulsion □□□□□□□ energy battery□□□ □□□□□□ motor□□□□□□□ regenerative braking □□□□□□□ motor□ □□□□□□ battery□□□□□□□□□ □□□□□□□□.

Study points

- Draw the energy path from battery to wheel and the reverse path during regeneration.
- Explain the different roles of the inverter, DC-DC converter, charger, and vehicle control unit.
- Compare the additional components required by an HEV.

Engineering / workplace application: The same energy-flow reasoning is used when diagnosing abnormal range, inverter temperature, motor torque, or battery state-of-charge behaviour.

Illustrative example: For a BEV, battery DC is converted by the inverter into controlled three-phase AC for a traction motor; the reduction gear converts motor speed and torque to wheel speed and tractive torque.

Common mistake: The onboard charger is not the same as the traction inverter; one primarily converts charging power while the other controls motor power during driving.

M1.03 — Configurations of Electric Vehicles (3 hours)

Concept and explanation

The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An FCEV uses a fuel cell to generate electricity, usually with a battery or buffer storage. An HEV combines an engine and an electrical path, while a PHEV adds a larger rechargeable battery and external charging capability. EVs may also be classified by whether torque is transmitted through a differential or through separate in-wheel or near-wheel drives.

> **Malayalam support:** BEV, FCEV, HEV, PHEV □□□□□ □□□□□□ energy source, charging method, engine presence, drive arrangement □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□. Differential □□□□□□□□ in-wheel drive □ □□□ wheel-□□□ □□□□□□□□□ motor control □□□□□□□□□□□□.

Study points

- Prepare a comparison using energy source, external charging, engine presence, and typical operating mode.
- Explain the meaning of with-differential and without-differential classifications.
- Use the official abbreviations correctly in answers.

> **Engineering / workplace application:** Vehicle configuration affects packaging, unsprung mass, control complexity, service procedures, and energy efficiency.

> **Illustrative example:** A BEV has no combustion engine for normal propulsion, whereas a PHEV can operate in electric mode and later use its engine when battery energy or power demand requires it.

> **Common mistake:** PHEV is not simply another name for BEV; a PHEV retains an engine and can normally be externally charged.

M1.04 — Types of hybrid electric vehicle (4 hours)

Concept and explanation

In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split device. A complex hybrid can include multiple motors, clutches, or flexible power paths. An Extended Range Electric Vehicle (EREV) primarily uses the electric drive for wheel propulsion while an engine-generator extends the available range when required.

> **Malayalam support:** Series HEV ▯ engine generator ▯▯▯▯▯▯▯▯▯, motor wheels ▯▯▯▯▯▯▯▯▯. Parallel HEV ▯ engine▯▯▯ motor▯▯▯ wheel torque ▯▯▯▯▯. Series-parallel power split device ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯. EREV-▯ engine ▯▯▯▯▯▯▯▯▯▯ range extender ▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯.

Study points

- Trace mechanical and electrical power paths for series, parallel, and series-parallel HEVs.
- State one advantage and one limitation of each major hybrid arrangement.
- Distinguish an EREV range-extender role from direct mechanical engine drive.

> **Engineering / workplace application:** Powertrain architecture is selected according to required acceleration, cruising efficiency, packaging, cost, and control strategy.

> **Illustrative example:** In a series HEV, the engine can run near an efficient operating point to generate electricity while the traction motor responds directly to wheel torque demand.

> **Common mistake:** Calling a series HEV a direct mechanical engine-to-wheel system reverses its defining power-flow arrangement.

> **Module summary:** EV and HEV classification becomes clear when the student follows the energy source, conversion path, and wheel-torque path.

OFFICIAL MODULE 2

EV Control Systems, Drive Trains and Braking

This module connects vehicle requirements to drive-system sizing, control, drive-train architecture, and braking. The central idea is coordinated power flow between energy storage, electronics, machines, wheels, and auxiliary systems.

Hours: 15

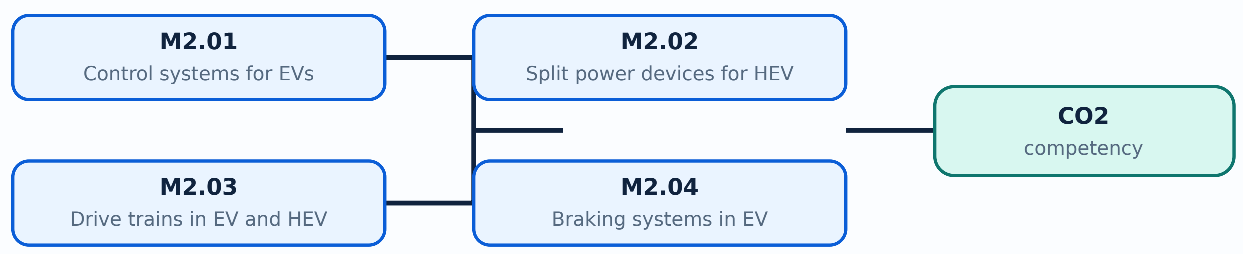
CO2 · Bloom level: Understanding

Learning goals

- List the principal functions of an EV control system and supporting subsystem.
- Compare power-split devices and single-, multi-, and in-wheel motor drive trains.
- Explain friction, regenerative, and hybrid braking at a system level.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Control systems for EVs (4 hours)

Concept and explanation

An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-electronics rating, energy-storage selection, communications network, and supporting subsystems such as cooling, steering, lighting, and braking. The controller must enforce voltage, current, temperature, and isolation limits while meeting the requested vehicle torque.

Malayalam support: EV control system accelerator input, battery state-of-charge, torque, temperature, charging, regenerative braking □□□□□□ □□□□□□□□□□□□□□□□. Sizing □□□□□□□□□□□ motor, power electronics, energy storage, communication, cooling □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Separate energy capacity requirements from peak power requirements.
- Identify feedback signals and protection limits used by a vehicle controller.
- Explain why motor, inverter, battery, and thermal sizing cannot be designed independently.

Engineering / workplace application: Battery-management and vehicle-control systems are central to safe charging, predictable range, and coordinated torque in production EVs.

Illustrative example: If the driver requests acceleration, the controller checks battery and temperature limits, commands inverter current, and reduces torque when a protection threshold is reached.

Common mistake: Selecting a large battery alone does not guarantee high acceleration; peak current, inverter power, motor torque, and thermal limits also matter.

M2.02 — Split power devices for HEV (3 hours)

Concept and explanation

A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator and motor torque so the engine can operate efficiently while the vehicle meets its speed and acceleration demand.

Malayalam support: Power-split device engine, generator, motor, wheels □□□□□□□□□□□□□□□□□□□ power flow □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□ □□□□□□□□. Planetary gear set □ speed-torque relationship □□□□□□□ engine efficient region □ □□□□□□□□□□□□□□□□□□□□□.

Study points

- Name the purpose of a power-split device.
- Explain that speed and torque are coupled through the gear arrangement.
- Relate generator control to engine operating-point management.

Engineering / workplace application: Power-split systems are used in hybrid transmissions to blend mechanical and electrical power during launch, cruising, and regeneration.

Illustrative example: During low-speed launch, the motor can provide wheel torque while the engine remains off or operates through the generator path.

Common mistake: A power-split device is not itself a battery; it is a mechanical/electromechanical arrangement that controls paths of power.

M2.03 — Drive trains in EV and HEV (4 hours)

Concept and explanation

An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery, fuel-cell, hybrid, or multiple-source power. Single-motor drives are simpler; multi-motor drives can distribute torque between axles or wheels. In-wheel drives reduce mechanical transmission parts but increase wheel mass and expose the motor to harsh conditions.

Malayalam support: Drive train □ source, converter, motor, reduction gear, differential □□□□□□□□□□ wheel motors, wheels □□□□□□ □□□□□□□□□□□□. Single motor □□□□□□□□; multi-motor torque distribution □□□□□□□□. In-wheel motor transmission parts □□□□□□□□□□□, □□□□□ unsprung mass □□□□□.

Study points

- Compare single-motor, multi-motor, and in-wheel drive arrangements.
- Explain how the power source changes the overall drive-train architecture.
- Mention packaging, control, efficiency, and service trade-offs.

Engineering / workplace application: Drive-train choice influences traction control, vehicle stability, packaging, and maintenance in buses, cars, and industrial vehicles.

Illustrative example: An all-wheel-drive EV can command different axle torques to improve traction on a low-friction surface.

Common mistake: More motors do not automatically mean better efficiency; converter losses, mass, control complexity, and operating point must be considered.

M2.04 — Braking systems in EV (4 hours)

Concept and explanation

EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provide the remaining deceleration and hold the vehicle at rest. Hybrid braking coordinates pedal feel, motor torque, hydraulic pressure, ABS, and stability control. The braking system must remain effective when the battery is full, cold, overheated, or unable to accept regeneration.

Malayalam support: Regenerative braking energy battery□□□□□□□□□ □□□□□□□□, □□□□□ battery full/cold/temperature limit □□□□□□□□□ friction brake □□□□□□□□□. Hybrid braking motor torque, hydraulic pressure, ABS, stability control □□□□□□ coordinate □□□□□□□□□□.

Study points

- Explain why regenerative braking cannot provide all braking in every condition.
- Differentiate regenerative, friction, and blended or hybrid braking.
- Relate braking control to ABS and battery acceptance limits.

Engineering / workplace application: Blended braking improves efficiency while maintaining reliable stopping distance in road vehicles.

Illustrative example: During a gentle stop with available battery capacity, the controller may request generator torque first and add hydraulic braking if the requested deceleration is not achieved.

Common mistake: Assuming regeneration is available during every braking event can lead to unsafe design and incorrect energy estimates.

Module summary: Control, drive-train, and brake decisions are coupled: the vehicle must satisfy torque and safety requirements while managing energy flow.

OFFICIAL MODULE 3

EV Design Requirements, Dynamics and Motors

This module gives the design language for vehicle range, speed, acceleration, resistive forces, tractive effort, transmission, energy consumption, and motor selection.

Hours: 15

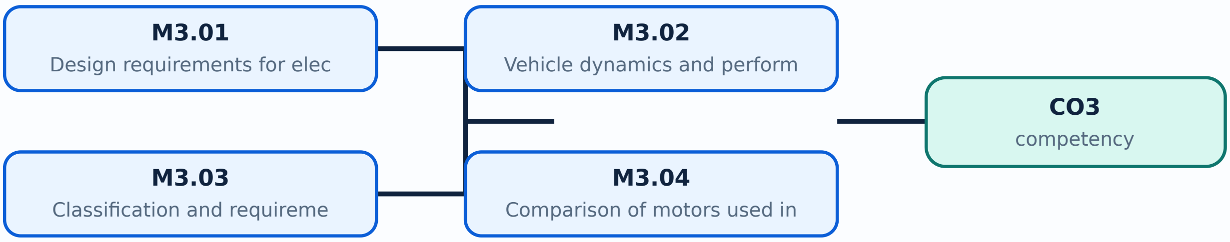
CO3 · Bloom level: Remembering / Understanding

Learning goals

- List the main performance requirements used in EV design.
- Calculate or identify rolling, aerodynamic, grade, and acceleration components of tractive effort.
- Classify EV motors and compare brushed and brushless DC motors with other practical choices.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Design requirements for electric vehicles (4 hours)

Concept and explanation

Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy storage can increase mass, while higher acceleration demands higher peak motor and inverter power. A design statement should define the duty cycle, road conditions, payload, ambient temperature, and allowable state-of-charge window before component ratings are selected.

Malayalam support: EV design □ range, maximum velocity, acceleration, gradeability, power, mass, charging time, safety, reliability □□□□□□ □□□□□□□□□□. Battery capacity □□□□□□□□□□□ mass □□□□□; □□□□□□ requirements □□□□□□□□□□ trade-off □□□□□□□□.

Study points

- Differentiate range, peak power, continuous power, and gradeability requirements.
- State the operating conditions that must be fixed before rating components.
- Explain the trade-off between battery mass and range.

Engineering / workplace application: Fleet duty-cycle data is used to size batteries and motors for buses, delivery vans, and industrial vehicles.

Illustrative example: A city bus with frequent stops may prioritise high regenerative capability and peak torque, while a highway vehicle may prioritise continuous power and aerodynamic efficiency.

Common mistake: Using only maximum speed to size an EV ignores launch torque, hill climbing, payload, thermal duty, and range.

M3.02 — Vehicle dynamics and performance (4 hours)

Concept and explanation

The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is F_trac = F_roll + F_aero + F_grade + m a. Rolling resistance may be approximated by C_rr m g; aerodynamic drag by 0.5 rho C_d A v^2; grade force by m g sin(theta). Wheel power is approximately P_wheel = F_trac v, and battery power is higher after allowing for transmission and electrical losses.

Malayalam support: Vehicle move ▯▯▯▯▯▯▯ rolling resistance, aerodynamic drag, grade resistance, acceleration resistance ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯. Speed ▯▯▯▯▯▯▯▯▯ aerodynamic drag v^2 ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯. Hill ▯▯▯▯▯▯▯▯▯ grade force ▯▯▯▯▯▯▯▯▯▯▯.

Study points

- Define each term in the tractive-force relation and give SI units.
- Explain why aerodynamic drag becomes significant at higher speed.
- Include transmission and electrical efficiency when moving from wheel power to battery power.

Engineering / workplace application: The same force balance supports route simulation, motor sizing, energy prediction, and regenerative-braking estimation.

Illustrative example: For a 1000 kg vehicle on a 5 degree slope, the grade component is approximately m g sin(5°), about 855 N before other resistances are added.

Common mistake: Mixing vehicle mass in kilograms with force in newtons without multiplying by g gives an incorrect resistance.

M3.03 — Classification and requirements of EV motors (4 hours)

Concept and explanation

EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, induction, permanent-magnet synchronous, and switched-reluctance families. Motor selection depends on torque-speed profile, inverter compatibility, cooling, fault tolerance, rare-earth dependence, and service requirements.

Malayalam support: EV motor □□ starting torque, wide speed range, high efficiency, regenerative operation, compact size, reliability □□□□□□ □□□□. Motor family □□□□□□□□□□□□□□□□□□□ torque-speed curve, inverter, cooling, cost, materials □□□□□□□ □□□□□□□□□□□□.

Study points

- List performance requirements of a traction motor.
- Classify motors by commutation or magnetic-field principle.
- Relate torque-speed characteristics to vehicle launch and cruising.

Engineering / workplace application: Traction motors are selected for the complete drive cycle rather than a single laboratory rating point.

Illustrative example: A motor can provide high low-speed torque for launch and then enter a constant-power region using field weakening or suitable control.

Common mistake: A high peak torque rating alone is not enough if the motor cannot dissipate heat during sustained hill climbing.

M3.04 — Comparison of motors used in EVs (3 hours)

Concept and explanation

Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent magnets and can tolerate high speed, but their control and efficiency characteristics differ. Permanent-magnet motors offer high power density but depend on magnets and careful thermal control. Switched-reluctance motors are robust and magnet-free but can have torque ripple and acoustic noise.

Malayalam support: Brushed DC motor maintenance □□□□□□ □□□; BLDC electronic commutation □□□□□□□□□□□□□□. Induction motor magnets □□□□□□□ □□□□□□□□□□□□□□. Permanent-magnet motor power density □□□□□□□□; SR motor robust □□□□□□□□□□ torque ripple □□□□□□□□.

Study points

- Compare commutation, maintenance, power density, control, cost, and typical limitations.
- Use a table in examination answers when comparing motor types.
- State that the best motor depends on the vehicle duty cycle and system constraints.

Engineering / workplace application: Motor comparison supports the selection of traction machines for two-wheelers, passenger vehicles, buses, and industrial drives.

Illustrative example: For a compact vehicle requiring high power density, a permanent-magnet motor may be attractive; for a cost-sensitive design, an induction or switched-reluctance alternative may be evaluated.

Common mistake: Saying that one motor type is universally best ignores application, duty cycle, cost, and control constraints.

Module summary: An EV design is a chain from performance requirement to force balance, power/energy requirement, transmission, and motor selection.

OFFICIAL MODULE 4

EV Braking, Sensors, Cooling, Service and Safety

This module focuses on the support systems that make an EV safe and usable: brake control, thermal management, sensors, maintenance, and electronic stability.

Hours: 15

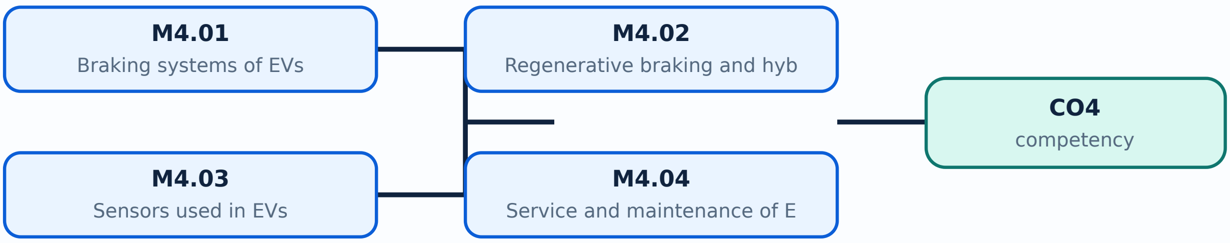
CO4 · Bloom level: Understanding

Learning goals

- Describe braking, air-conditioning, adaptive cruise control, and ABS-related requirements.
- Explain regenerative and hybrid braking together with battery and motor cooling.
- Identify sensors and outline safe maintenance of EV subsystems.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Braking systems of EVs (3 hours)

Concept and explanation

An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-conditioning loads affect electrical energy consumption, while adaptive cruise control uses sensors and control logic to maintain speed or distance and can request propulsion or braking torque.

Malayalam support: EV brake system □ brake input, controller, actuator, friction brake, regenerative path, wheel-speed sensor □□□□□□ □□□□□□□□. ABS wheel lock □□□□□□□□□□ pressure modulate □□□□□□□. Air-conditioning energy consumption-□□ □□□□□□□□□.

Study points

- State the purpose of ABS and adaptive cruise control.
- Identify brake-system elements that must remain available under electrical faults.
- Relate auxiliary loads such as air conditioning to EV range.

Engineering / workplace application: Brake-by-wire and stability systems coordinate vehicle safety with energy recovery without compromising stopping performance.

Illustrative example: If a wheel-speed sensor detects incipient lock, ABS reduces and reapplies pressure while the vehicle controller manages any regenerative contribution.

Common mistake: Adaptive cruise control is not a substitute for driver attention or an independent guarantee of safe stopping.

M4.02 — Regenerative braking and hybrid braking (4 hours)

Concept and explanation

Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control system considers battery state of charge, motor speed, temperature, tyre adhesion, ABS demand, and brake-pedal feel. Regeneration is reduced when the battery cannot accept charge or when strong emergency braking is required.

Malayalam support: Motor generator □□□ □□□□□□□□□□□□□□□□□□□ kinetic energy electrical energy □□□□□ battery□□□□□□□□ □□□□□□□□. Battery limit, tyre grip, ABS demand □□□□□□ □□□□□ friction brake □□□□ □□□□□□□□□□ blended braking □□□□□□□□□□.

Study points

- Explain the energy conversion during regeneration.
- List conditions that reduce regenerative braking.
- Describe how brake-pedal feel and stopping safety are preserved.

Engineering / workplace application: Regeneration is particularly useful in stop-start urban driving, downhill operation, and fleet vehicles.

Illustrative example: When the battery is near full charge, the controller reduces generator torque and relies more on friction braking while maintaining the requested deceleration.

Common mistake: Regenerative braking should not be treated as free energy; conversion losses and battery limits remain.

M4.03 — Sensors used in EVs (4 hours)

Concept and explanation

Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protection, and charger monitoring. A coolant-level sensor detects insufficient coolant that could cause thermal damage. Sensor signals are filtered, plausibility-checked, and used by the vehicle controller for warnings, derating, or shutdown.

Malayalam support: Temperature sensor, MEMS, current sensor, coolant-level sensor ▯▯▯▯▯▯ battery, motor, inverter, cooling system ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. Sensor signal fault ▯▯▯▯▯▯▯▯ controller warning ▯▯▯▯▯▯▯▯▯▯ derating ▯▯▯▯▯▯▯.

Study points

- Match each sensor with its measured quantity and system use.
- Explain why redundant or plausibility checks are important for safety signals.
- Distinguish measurement, signal conditioning, and controller action.

Engineering / workplace application: Sensors are used in battery-management systems, motor drives, thermal control, driver assistance, and electronic stability systems.

Illustrative example: If coolant temperature rises above a calibrated limit, the controller can reduce power, command cooling, store a diagnostic code, and alert the driver.

Common mistake: A sensor reading should not be trusted blindly; open-circuit, short-circuit, out-of-range, and implausible values require diagnosis.

M4.04 — Service and maintenance of EV components (4 hours)

Concept and explanation

EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, battery and cooling-system checks, motor and inverter inspection, brake and tyre service, sensor checks, electronic-stability diagnostics, and safe handling of charging equipment. Only trained personnel should work on high-voltage circuits using approved isolation, PPE, insulated tools, and lockout procedures.

Malayalam support: EV maintenance-□ engine oil/exhaust service □□□□□□□□□□ high-voltage safety □□□□□□□□□□□. Battery, cooling, motor, inverter, brakes, tyres, sensors, charging equipment □□□□□□ □□□□□□□□□□□□. Trained person □□□□□□ HV circuit service □□□□□□□.

Study points

- Compare EV maintenance with combustion-engine maintenance.
- List high-voltage safety steps before service.
- Include battery, motor, brakes, sensors, cooling, and electronic stability in a maintenance plan.

Engineering / workplace application: Service procedures are applied in workshops, fleet depots, charging stations, and accident-recovery situations.

Illustrative example: A safe service sequence is identify the vehicle, switch off, isolate and lock out the high-voltage source, wait the specified discharge time, verify absence of voltage, then begin work.

Common mistake: Disconnecting a visible connector does not by itself prove that a high-voltage system is safe; isolation must be verified according to procedure.

Module summary: Reliable EV operation depends on coordinated brakes, sensors, cooling, maintenance, and safety procedures, not only on the traction motor.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

learning-map diagram.

Module 2: EV Control

rning-map diagram.

Module 3: EV Design

rning-map diagram.

Module 4: EV Braking,

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Source anomaly: The supplied course outline lists Series Test I and Series Test II as 1 hour within module entries, while the Course Outcomes table separately lists Series Test as 2 hours.
- Source anomaly: The module table gives Module 4 a total of 15 hours, while CO4 states 13 hours; the 1-hour Series Test II is also shown under Module 4. These official values are preserved without silent correction.
- No separate CIE/ESE mark distribution is stated in the supplied PDF.

Module-wise Questions and Answers

module-1 — Electric and Hybrid Vehicle Fundamentals

1. Explain Need for hybrid and electric vehicles. [3 marks]

Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and the possibility of using electricity from diverse sources. Historical development moved from early battery vehicles through modern power electronics, lithium-based batteries, charging infrastructure, and electronically controlled propulsion. Social importance includes quieter transport and reduced local pollution; environmental benefit depends on the electricity-generation mix, battery manufacture, and responsible end-of-life management.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Components and working principles of electric and hybrid vehicles. [3 marks]

A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loads, and vehicle control unit. An HEV adds an engine, fuel system, generator or motor-generator, and a supervisory energy-management controller. During propulsion, stored electrical energy passes through the inverter to the motor; during regenerative deceleration, the power flow reverses and the motor operates as a generator.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Configurations of Electric Vehicles. [3 marks]

The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An FCEV uses a fuel cell to generate electricity, usually with a battery or buffer storage. An HEV combines an engine and an electrical path, while a PHEV adds a larger rechargeable battery and external charging capability. EVs may also be classified by whether torque is transmitted through a differential or through separate in-wheel or near-wheel drives.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Types of hybrid electric vehicle. [3 marks]

In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split device. A complex hybrid can include multiple motors, clutches, or flexible power paths. An Extended Range Electric Vehicle (EREV) primarily uses the electric drive for wheel propulsion while an engine-generator extends the available range when required.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — EV Control Systems, Drive Trains and Braking

5. Explain Control systems for EVs. [3 marks]

An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-electronics rating, energy-storage selection, communications network, and supporting subsystems such as cooling, steering, lighting, and braking. The controller must enforce voltage, current, temperature, and isolation limits while meeting the requested vehicle torque.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Split power devices for HEV. [3 marks]

A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator and motor torque so the engine can operate efficiently while the vehicle meets its speed and acceleration demand.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Drive trains in EV and HEV. [3 marks]

An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery, fuel-cell, hybrid, or multiple-source power. Single-motor drives are simpler; multi-motor drives can distribute torque between axles or wheels. In-wheel drives reduce mechanical transmission parts but increase wheel mass and expose the motor to harsh conditions.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Braking systems in EV. [3 marks]

EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provide the remaining deceleration and hold the vehicle at rest. Hybrid braking coordinates pedal feel, motor torque, hydraulic pressure, ABS, and stability control. The braking system must remain effective when the battery is full, cold, overheated, or unable to accept regeneration.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — EV Design Requirements, Dynamics and Motors

9. Explain Design requirements for electric vehicles. [3 marks]

Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy storage can increase mass, while higher acceleration demands higher peak motor and inverter power. A design statement should define the duty cycle, road conditions, payload, ambient temperature, and allowable state-of-charge window before component ratings are selected.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Vehicle dynamics and performance. **[3 marks]**

The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is F_trac = F_roll + F_aero + F_grade + m a. Rolling resistance may be approximated by C_rr m g; aerodynamic drag by 0.5 rho C_d A v^2; grade force by m g sin(theta). Wheel power is approximately P_wheel = F_trac v, and battery power is higher after allowing for transmission and electrical losses.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Classification and requirements of EV motors. **[3 marks]**

EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, induction, permanent-magnet synchronous, and switched-reluctance families. Motor selection depends on torque-speed profile, inverter compatibility, cooling, fault tolerance, rare-earth dependence, and service requirements.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Comparison of motors used in EVs. **[3 marks]**

Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent magnets and can tolerate high speed, but their control and efficiency characteristics differ. Permanent-magnet motors offer high power density but depend on magnets and careful thermal control. Switched-reluctance motors are robust and magnet-free but can have torque ripple and acoustic noise.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — EV Braking, Sensors, Cooling, Service and Safety

13. Explain Braking systems of EVs. [3 marks]

An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-conditioning loads affect electrical energy consumption, while adaptive cruise control uses sensors and control logic to maintain speed or distance and can request propulsion or braking torque.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Regenerative braking and hybrid braking. [3 marks]

Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control system considers battery state of charge, motor speed, temperature, tyre adhesion, ABS demand, and brake-pedal feel. Regeneration is reduced when the battery cannot accept charge or when strong emergency braking is required.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Sensors used in EVs. [3 marks]

Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protection, and charger monitoring. A coolant-level sensor detects insufficient coolant that could cause thermal damage. Sensor signals are filtered, plausibility-checked, and used by the vehicle controller for warnings, derating, or shutdown.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Service and maintenance of EV components. **[3 marks]**

EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, battery and cooling-system checks, motor and inverter inspection, brake and tyre service, sensor checks, electronic-stability diagnostics, and safe handling of charging equipment. Only trained personnel should work on high-voltage circuits using approved isolation, PPE, insulated tools, and lockout procedures.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need for hybrid and electric vehicles* with one relevant application. (5 marks)
2. **2.** Define or explain *Components and working principles of electric and hybrid vehicles* with one relevant application. (5 marks)
3. **3.** Define or explain *Control systems for EVs* with one relevant application. (5 marks)
4. **4.** Define or explain *Split power devices for HEV* with one relevant application. (5 marks)
5. **5.** Define or explain *Design requirements for electric vehicles* with one relevant application. (5 marks)
6. **6.** Define or explain *Vehicle dynamics and performance* with one relevant application. (5 marks)
7. **7.** Define or explain *Braking systems of EVs* with one relevant application. (5 marks)
8. **8.** Define or explain *Regenerative braking and hybrid braking* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Electric and Hybrid Vehicle Fundamentals:** EV and HEV classification becomes clear when the student follows the energy source, conversion path, and wheel-torque path.
- **EV Control Systems, Drive Trains and Braking:** Control, drive-train, and brake decisions are coupled: the vehicle must satisfy torque and safety requirements while managing energy flow.
- **EV Design Requirements, Dynamics and Motors:** An EV design is a chain from performance requirement to force balance, power/energy requirement, transmission, and motor selection.
- **EV Braking, Sensors, Cooling, Service and Safety:** Reliable EV operation depends on coordinated brakes, sensors, cooling, maintenance, and safety procedures, not only on the traction motor.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Need for hybrid and electric vehicles	Electric vehicles use an electric propulsion system instead of, or in combination with, an internal-combustion engine. The need for EVs and HEVs arises from energy efficiency, reduced tail-pipe emissions, urban air-quality concerns, regenerative braking, and t
Components and working principles of electric and hybrid vehicles	A battery-electric vehicle normally contains a traction battery, battery-management system, contactors and protection, inverter or motor controller, traction motor, reduction gear, differential or individual wheel drive, charger, DC-DC converter, auxiliary loa
Configurations of Electric Vehicles	The syllabus identifies Pure Electric Vehicle (PE), Battery Electric Vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV). A BEV obtains its traction energy from a rechargeable battery. An
Types of hybrid electric vehicle	In a series HEV, the engine drives a generator and the electric motor drives the wheels. In a parallel HEV, both engine and motor can contribute mechanical torque to the drivetrain. A series-parallel or power-split HEV combines both paths using a power-split d
Control systems for EVs	An EV control system supervises accelerator demand, torque production, battery state of charge, thermal limits, charging, regenerative braking, traction, communication, and safety interlocks. Sizing the drive system requires a propulsion-motor rating, power-el
Split power devices for HEV	A power-split device divides or combines power between the engine, generator, motor, and wheels. In a planetary gear arrangement, sun, planet, and ring members can provide different speed and torque relationships. The supervisory controller changes generator a
Drive trains in EV and HEV	An electric drive train includes the source, power converter, electric machine, reduction transmission, differential or wheel motors, and wheels. General configurations may be front-wheel, rear-wheel, or all-wheel drive. Alternatives are also based on battery,
Braking systems in EV	EV braking blends regenerative braking with friction braking. When the motor operates as a generator, kinetic energy is converted into electrical energy and returned to the storage system, subject to battery acceptance and safety limits. Friction brakes provid
Design requirements for electric vehicles	Important EV design requirements include driving range, maximum velocity, acceleration time, gradeability, required wheel power, vehicle mass, charging time, packaging, safety, reliability, and energy consumption. The requirements are linked: larger energy sto
Vehicle dynamics and performance	The tractive force at the tyre contact patch must overcome rolling resistance, aerodynamic drag, grade resistance, and acceleration resistance. A useful introductory relation is F_trac = F_roll + F_aero + F_grade + m a. Rolling resistance may be approximated b
Classification and requirements of EV motors	EV motors require high starting torque, wide speed range, high efficiency over the duty cycle, regenerative capability, controllability, compact size, low noise, reliability, and manageable cost. Common classifications include brushed DC, brushless DC, inducti

Term	Short meaning
Comparison of motors used in EVs	Brushed DC motors have simple control and high starting torque but require brush and commutator maintenance. Brushless DC motors remove mechanical brushes, improve efficiency and reliability, and require electronic commutation. Induction motors avoid permanent
Braking systems of EVs	An EV braking system includes the driver input, brake controller, hydraulic or electromechanical actuator, friction brakes, regenerative path, wheel-speed sensors, and safety monitoring. ABS prevents excessive wheel slip by modulating brake pressure. Air-condi
Regenerative braking and hybrid braking	Regenerative braking converts a portion of vehicle kinetic energy into electrical energy through the traction motor operating as a generator. Hybrid or blended braking combines this torque with friction braking to meet the requested deceleration. The control s
Sensors used in EVs	Temperature sensors protect cells, motors, inverters, and coolant circuits. MEMS sensors can measure acceleration, pressure, or other physical quantities in compact packages. Current sensors support torque control, battery state estimation, over-current protec
Service and maintenance of EV components	EV maintenance differs from combustion-engine maintenance because there is no engine oil, exhaust, or conventional fuel system, but high-voltage isolation and electrical safety become essential. Maintenance includes visual inspection, isolation verification, b

Official References and Resources

References listed in the supplied syllabus

1. Jack Erjavec and Jeff Arias, Hybrid, Electric and Fuel-cell Vehicles, Cengage Learning India Pvt Ltd., New Delhi.
2. S. Onori, L. Serrao and G. Rizzoni, Hybrid Electric Vehicles: Energy Management Strategies, Springer, 2015.
3. T. Denton, Electric and Hybrid Vehicles, Routledge.
4. A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi.
5. C. Mi, M. A. Masrur and D. W. Gao, Hybrid Electric Vehicles: Principles and Applications with ... [the supplied PDF truncates the title here].

Official learning resources listed

- https://nptel.ac.in/courses/108/103/108103009/
- https://nptel.ac.in/courses/108/106/108106170
- https://www.sciencedirect.com/science/article/pii/S2095756420300647
- https://www.metisengineering.com

- https://onlinecourses.nptel.ac.in/noc20_ee99/preview

In-page Search